

ME2801, Lab #1, USV Modeling and System Identification

1 Learning Objectives

By the end of this lab you should be able to:

Derive Linear transfer function models for a marine vessel from first principles (Newton's second law).

Identify The physical meaning of the DC gain and time constant of a first-order plant.

Describe How propeller thrust controls surge speed and how rudder moment controls heading.

Simulate The **open-loop** step response for nominal USV parameters, to predict the dynamics of speed and yaw response to manual inputs.

Collect Collect experimental step-response data from a USV running ArduPilot and plot the open-loop response, measuring step-response metrics: rise time, settling time, and percent overshoot.

System Identification Fit the model to the data by selecting transfer function parameters so that the model output matches the measured step response.

2 Deliverable

You will work in small teams on this lab. Each team will create a short document with the following elements. Each member of the team will submit a copy of the document (with all the team members names) through Sakai.

- Title and names of team members.
- Nominal Response: Figures for the open-loop step response for surge and yaw-rate using the provided nominal parameters. Include a narrative that compares the expected response metrics (rise-time, overshoot, steady-state error) with the observed metrics.
- Experimental Summary: Figures of the experimental results for surge and yaw-rate. This is solely based on the data. Include a narrative to identify the empirical system parameters for each response: time-constant and DC gain.
- System Identification: After completing the system identification (via curve fitting), present figures with experimental open-loop response (the data) and the identified transfer function response (the model) on the same axes. Include a narrative to quantitatively compare the model to the experiment with metrics such as rise time, settling time, and percent overshoot. Include a table for a side-by-side comparison of experimental and model metrics.

The *narratives* should be brief to explain the key associated figures and metrics. The narratives should not include the process, just the results.

3 Physical System

3.1 Vehicle Description

The vehicle is a small, single-hull unmanned surface vehicle (USV) approximately 1.1 m in length and 5 kg in mass. It carries one fixed-pitch propeller on the centerline for thrust and one stern rudder for steering. Because the propeller is on the centerline and because we are using the model for control, we can consider thrust and steering to be effectively decoupled — thrust controls forward speed; the rudder controls heading direction.

3.2 Coordinate Frame and Degrees of Freedom

We adopt the standard SNAME body-fixed frame. The x -axis points forward (bow), the y -axis points to port, and the z -axis points upward. For this lab we model two scalar motions:

DOF	Variable	Units	Description
Surge	u	m/s	Forward (longitudinal) velocity along body x -axis
Yaw	r, ψ	rad/s, rad	Yaw rate $r = \dot{\psi}$; heading angle ψ measured from North

Sway (lateral velocity) and roll are neglected. For a slow, narrow-beam hull this is a reasonable simplification. The two channels (surge and yaw) are treated as independent and are controlled by separate PID loops.

4 Model Derivation

4.1 Surge Channel

4.1.1 Equation of Motion

Applying Newton's second law in the surge (forward) direction:

$$m_{\text{eff}} \dot{u} = F_{\text{thrust}} - F_{\text{drag}} \quad (1)$$

where m_{eff} is an effective inertia that includes the vessel's physical mass m plus an added-mass correction m_a due to the entrained water that accelerates with the hull:

$$m_{\text{eff}} = m + m_a \approx 1.20 m$$

For the drag force we adopt a *linear* (viscous) model. In reality, hydrodynamic drag is approximately quadratic in velocity ($F_{\text{drag}} \propto u^2$), but linearizing about a nominal operating speed u_0 gives:

$$F_{\text{drag}} = b_u u$$

where b_u is the linearized surge drag coefficient [N·s/m]. Absorbing the 1.20 factor into the lumped parameter m , the complete surge equation of motion is:

$$\boxed{m \dot{u} + b_u u = F} \quad (2)$$

4.1.2 Transfer Function

Taking the Laplace transform of (2) with zero initial conditions:

$$m s U(s) + b_u U(s) = F(s) \quad \implies \quad U(s) (ms + b_u) = F(s)$$

The surge plant transfer function — from thrust F [N] to surge speed U [m/s] — is:

$$G_{\text{surge}}(s) = \frac{U(s)}{F(s)} = \frac{1}{ms + b_u} = \frac{K_u}{\tau_u s + 1} \quad (3)$$

This is a **first-order system** with:

- **DC gain:** $K_u = 1/b_u$ [m s⁻¹/N] — steady-state speed per unit thrust
- **Time constant:** $\tau_u = m/b_u$ [s] — time to reach 63% of final speed
- **Pole:** $s = -b_u/m$ (left-half plane — open-loop stable)

4.1.3 Command-Scaled Transfer Function

In practice the propeller is driven by a speed controller that accepts a thrust command $\delta_T \in [0, 100]$ expressed as a percentage of full thrust. Assuming a linear actuator model:

$$F = F_{\max} \frac{\delta_T}{100} \quad (4)$$

where $F_{\max}$ is the maximum thrust produced at 100% command. Substituting (4) into (2) and taking the Laplace transform gives the command-scaled surge transfer function:

$$G_{\text{surge}}^*(s) \equiv \frac{U(s)}{\Delta_T(s)} = \frac{F_{\max}/100}{ms + b_u} = \frac{K_u^*}{\tau_u s + 1} \quad (5)$$

where the command-scaled DC gain and time constant are:

$$K_u^* = \frac{F_{\max}}{100 b_u} \left[\frac{\text{m/s}}{\%} \right], \quad \tau_u = \frac{m}{b_u} \text{ [s]}$$

The time constant τ_u is unchanged — it depends only on the vehicle dynamics, not on the actuator scaling. Only the DC gain absorbs the actuator gain $F_{\max}/100$. This means that from a step-response experiment, τ_u can be read directly from the response shape while K_u^* is determined from the steady-state speed achieved at a known command percentage.

4.2 Yaw Channel

4.2.1 Equation of Motion

Applying Newton's second law for rotation about the vertical (z) axis:

$$I_{\text{eff}} \dot{r} = N_{\text{rudder}} - N_{\text{drag}} \quad (6)$$

where $I_{\text{eff}} = I + I_a$ is the effective yaw inertia (hull inertia plus rotational added inertia), $r = \dot{\psi}$ is the yaw rate, N_{rudder} is the hydrodynamic moment produced by the rudder [N·m], and $N_{\text{drag}} = b_r r$ is a linear damping moment. The yaw-rate equation is:

$$\boxed{I \dot{r} + b_r r = N} \quad (7)$$

4.2.2 Heading Transfer Function

Because the commanded output is usually the heading angle ψ rather than yaw rate r , and since $r = \dot{\psi}$, we note that $\Psi(s) = R(s)/s$.

The yaw-rate plant transfer function (moment N to yaw rate R) is the same first-order form as surge:

$$G_r(s) = \frac{R(s)}{N(s)} = \frac{1}{Is + b_r}$$

The heading plant transfer function (moment N to heading angle Ψ) adds an integrator:

$$G_{\text{yaw}}(s) = \frac{\Psi(s)}{N(s)} = \frac{1}{s(Is + b_r)} = \frac{K_r/\tau_r}{s(\tau_r s + 1)} \quad (8)$$

The heading plant (8) has a **free integrator** (a pole at $s = 0$). Physically: a constant rudder moment produces a constantly growing heading angle. As a control-design consequence, even a proportional-only controller achieves zero steady-state error to a step heading command — verify this with the Final Value Theorem as part of your pre-lab.

4.2.3 Command-Scaled Transfer Function

The rudder servo accepts a displacement command $\delta_R \in [-100, 100]$ expressed as a percentage of full deflection, with positive values corresponding to a port (left) turn. Assuming a linear mapping from command to hydrodynamic moment:

$$N = N_{\text{max}} \frac{\delta_R}{100} \quad (9)$$

where N_{max} is the maximum yaw moment at full rudder deflection. Substituting (9) into (7) yields the command-scaled yaw-rate transfer function:

$$G_r^*(s) \equiv \frac{R(s)}{\Delta_R(s)} = \frac{N_{\text{max}}/100}{Is + b_r} = \frac{K_r^*}{\tau_r s + 1} \quad (10)$$

and the command-scaled heading transfer function:

$$G_{\text{yaw}}^*(s) \equiv \frac{\Psi(s)}{\Delta_R(s)} = \frac{N_{\text{max}}/100}{s(Is + b_r)} = \frac{K_r^*}{s(\tau_r s + 1)} \quad (11)$$

where:

$$K_r^* = \frac{N_{\text{max}}}{100 b_r} \left[\frac{\text{rad/s}}{\%} \right], \quad \tau_r = \frac{I}{b_r} \text{ [s]}$$

As with surge, the time constant τ_r is unaffected by actuator scaling. Note that because the heading plant contains a free integrator, a step rudder command produces a ramp in heading angle at

steady state — the vehicle turns continuously. System identification for the yaw channel is therefore performed on the *yaw-rate* response $r(t)$, which does reach a steady state, and the heading model is then constructed by appending the integrator.

5 Nominal Parameters

The following lumped parameters are used in simulation. They are representative of a ~ 5 kg, 1.1 m recreational-grade USV.

Symbol	Value	Units	Description
m	5.0	kg	Effective surge mass (hull + added mass)
b_u	8.0	N·s/m	Linear surge drag coefficient
I	0.8	kg·m ²	Effective yaw inertia (hull + added inertia)
b_r	1.2	N·m·s/rad	Linear yaw damping coefficient
τ_u	0.625	s	Surge time constant $\tau_u = m/b_u$
τ_r	0.667	s	Yaw-rate time constant $\tau_r = I/b_r$
K_u	0.125	(m/s)/N	Surge DC gain (physical input) $K_u = 1/b_u$
K_r	0.833	(rad/s)/(N·m)	Yaw-rate DC gain (physical input) $K_r = 1/b_r$
K_u^*	—	(m/s)/%	Surge DC gain (command-scaled) $K_u^* = F_{\max}/(100 b_u)$
K_r^*	—	(rad/s)/%	Yaw-rate DC gain (command-scaled) $K_r^* = N_{\max}/(100 b_r)$

The command-scaled gains K_u^* and K_r^* depend on $F_{\max}$ and $N_{\max}$, which are vehicle-specific and determined by system identification (Section 7). Dashes indicate that nominal values are not assigned a priori.

6 Transfer Function Summary

Physical-input forms (force/moment in N and N·m)

Name	Expression	Notes
Surge plant	$G_{\text{surge}}(s) = \frac{1}{ms + b_u}$	F [N] $\rightarrow u$ [m/s]; pole at $s = -b_u/m$
Yaw-rate plant	$G_r(s) = \frac{1}{Is + b_r}$	N [N·m] $\rightarrow r$ [rad/s]; pole at $s = -b_r/I$
Heading plant	$G_{\text{yaw}}(s) = \frac{1}{s(Is + b_r)}$	N [N·m] $\rightarrow \psi$ [rad]; poles at $0, -b_r/I$

Command-scaled forms (input in % of full actuation)

Name	Expression	Notes
Surge plant	$G_{\text{surge}}^*(s) = \frac{K_u^*}{\tau_u s + 1}$	$\delta_T [\%] \rightarrow u [\text{m/s}];$ $K_u^* = \frac{F_{\text{max}}}{100 b_u}$
Yaw-rate plant	$G_r^*(s) = \frac{K_r^*}{\tau_r s + 1}$	$\delta_R [\%] \rightarrow r [\text{rad/s}];$ $K_r^* = \frac{N_{\text{max}}}{100 b_r}$
Heading plant	$G_{\text{yaw}}^*(s) = \frac{K_r^*}{s(\tau_r s + 1)}$	$\delta_R [\%] \rightarrow \psi [\text{rad}]$

The parameters to be identified from experiment are K_u^* , τ_u , K_r^* , and τ_r . The time constants are shared between the physical and command-scaled forms; only the DC gains differ by the actuator scale factor.

7 System Identification from Step Responses

7.1 Overview

System identification is the process of determining model parameters from measured input–output data. For a first-order system the step response has a simple analytical form that makes parameter extraction straightforward. The procedure below applies to both the surge channel (input: δ_T ; output: u) and the yaw channel (input: δ_R ; output: r). In each case a *step command* is applied — the input is held at zero, then switched to a constant value δ_0 and held there — and the output is recorded.

7.2 Analytical Step Response of a First-Order System

For a generic first-order command-scaled plant

$$G^*(s) = \frac{K^*}{\tau s + 1}$$

the response to a step input of magnitude δ_0 (applied at $t = 0$, starting from rest) is:

$$y(t) = K^* \delta_0 (1 - e^{-t/\tau}), \quad t \geq 0 \quad (12)$$

Two key features of (12) are used for identification:

- **Steady-state value:** as $t \rightarrow \infty$, $y_\infty = K^* \delta_0$. Therefore:

$$K^* = \frac{y_\infty}{\delta_0} \quad (13)$$

- **Time constant:** at $t = \tau$ the output reaches 63.2% of its final value, i.e. $y(\tau) = 0.632 y_\infty$. Therefore τ is the time at which the measured response first crosses $0.632 y_\infty$:

$$\tau = t|_{y=0.632 y_\infty} \quad (14)$$

A complementary graphical check is the *initial slope*: at $t = 0$, $\dot{y}(0) = K^*\delta_0/\tau$, so a tangent line drawn at the origin intersects the steady-state value at $t = \tau$. This provides a second, independent estimate of τ .

7.3 Surge Model System Identification

The ArduPilot autopilot firmware should be configured to log all internal data to the removeable SD card. To check this, verify in MP that `LOG_BITMASK = 65535`.

1. Setup: Designate a pilot (driving the USV with R/C transmitter), a planner (to monitor in mission planner) and a scribe (taking notes with time stamps). For each experiment trial, have the scribe note the time and any key observations. This will make it much easier to sort through the data post-experiment.
2. Speed response trials:
 - Prior to each trial, the scribe should note the time and record any identifying behavior.
 - With the vehicle at rest on open water have the pilot apply a step thrust command $\delta_T = \delta_0$ (a value in the range 20–50% is recommended to remain in the approximately linear regime while producing a measurable response). Hold the command constant until the surge speed u reaches a clear steady state (typically $5\tau_u$ or about 10 s). When in doubt, hold the command longer to ensure you capture steady-state.
 - During the experiment, if the planner can record $u(t)$ (forward speed) and R/C command from mission planner, it will further help with post-processing.
 - Conduct 2-3 successful straight line surge tests.

7.4 Yaw Model System Identification

1. With the vehicle moving at a low, approximately constant surge speed (so that rudder authority is meaningful), apply a step rudder command $\delta_R = \delta_0$ and hold it constant. Record the *yaw rate* $r(t)$ from the IMU gyroscope.

7.4.1 Practical Notes

- Perform the yaw experiment at a fixed, repeatable surge speed. Rudder moment depends on flow velocity over the rudder, so $N_{\max}$ (and therefore K_r^*) will vary with speed. Note the surge speed at which identification was performed; the yaw model is only valid near that operating point.
- Use symmetric step commands ($+\delta_0$ then $-\delta_0$) and average the identified gains to reduce the effect of any asymmetry in the rudder or hull.

7.4.2 Verify Logged Data

ArduPilot logs data in a binary format called `.bin`. The files are stored on the SD card and can be downloaded to a computer for analysis. It is good practice to verify that the data is being logged correctly before leaving the field. We will discuss methods of doing this during the lab session.

7.5 System Identification: Fitting the Model to Data

7.5.1 Graphical Method

To get a first rough estimate of the parameters, plot the measured step response $y(t)$ and apply the formulas (13) and (14) to extract K^* and τ for each of the surge and yaw models. This is a quick way to get ballpark estimates without needing to do any curve fitting. We will also use these estimates as initial guesses for the more rigorous curve-fitting method described next.

Produce two separate figures. For the surge channel, plot $u(t)$ and annotate the steady-state value y_∞ and the time at which the response first crosses $0.632 y_\infty$. For the yaw channel, plot $r(t)$ and similarly annotate the steady-state value and the 63.2% crossing time. On these same axes, also plot the step response predicted by the identified parameters to visually check the fit.

7.5.2 Curve Fitting

Once initial estimates $\hat{K}^*$ and $\hat{\tau}$ have been obtained graphically, a better fit can be achieved by minimizing the least-squares error between the analytical step response (12) and the measured data. Given N data samples (t_k, y_k) , define the residual vector:

$$\varepsilon_k = y_k - \hat{K}^* \delta_0 (1 - e^{-t_k/\hat{\tau}})$$

and minimize $\sum_k \varepsilon_k^2$ over (K^*, τ) . In MATLAB this is accomplished with `lsqcurvefit`, which is included in the Optimization Toolbox. The code snippet below shows how to set up the optimization problem:

```
function y = first_order_step(params, t, delta0)
    K_star = params(1);
    tau = params(2);
    y = K_star * delta0 * (1 - exp(-t / tau));
end

% t_data, y_data: arrays from experiment; delta0: step magnitude
params0 = [K_hat, tau_hat]; % initial guesses from graphical method
params_fit = lsqcurvefit(...
    @(p, t) first_order_step(p, t, delta0), ...
    params0, t_data, y_data);
K_fit = params_fit(1);
tau_fit = params_fit(2);
```

The graphical estimates serve as initial guesses p_0 ; providing good starting values is important because the optimization is nonlinear in τ .

7.6 Validation

After fitting always validate the identified model against data not used in the fit. Create your transfer function models for surge and yaw-rate using the identified parameters from the curve fitting, and plot the model-predicted step response on top of the measured data. Check that:

- The model captures the key features of the response: rise time, settling time, and percent overshoot.
- Report the root-mean-square error:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{k=1}^N \epsilon_k^2}$$

as a quantitative measure of fit quality.

7.7 Summary of Identified Parameters

Record your identified parameters in the following table:

Channel	δ_0 [%]	y_∞	K^* (identified)	τ (identified)
Surge				
Yaw rate				

Units: K_u^* in (m/s)/%; K_r^* in (rad/s)/%; τ_u, τ_r in s.